MIL-P-46847B(MI)
21 November 1984
SUPERSEDING
MIL-P-46847A(MI)
2 August 1968

MILITARY SPECIFICATION

PLASTIC MATERIAL, FOAMED POLYURETHANE FOR ENCAPSULATING ELECTRONIC COMPONENTS

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers six types of semi-rigid, room temperature curing, foam-in-place polyurethane foam.

1.2 Classification. The polyurethane foam shall be furnished in the following types (see 6.2), as classified by their densities and compressive strengths:

Type I - 2 pounds per cubic foot, 8 pounds per square inch

Type II - 5 pounds per cubic foot, 20 pounds per square inch

Type III - 8 pounds per cubic foot, 100 pounds per square inch

Type IV - 12 pounds per cubic foot, 275 pounds per square inch

Type V - 18 pounds per cubic foot, 450 pounds per square inch

Type VI - 2 pounds per cubic foot, 40 pounds per square inch

* 1.3 Military part number. The military part number shall consist of the letter B, the basic number of this specification, plus dash numbers and letters denoting type, quantity, and container size.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-EDS, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

FSC 9330

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Example:

B 46847	- III	-*01 PT
Basic Mil Spec No.	Type (See 1.2 & Table I)	Quantity & Container Size PT -- Pint QT -- Quart GL -- Gallon

*First two digits denote numerical quantity; last two letters identify container size.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

*	QQ-S-781	Strapping, Steel, and Seals
	PPP-B-585	Boxes, Wood, Wirebound
	PPP-B-601	Boxes, Wood, Cleated-Plywood
	PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
	PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

	MIL-P-116	Preservation, Methods of
*	MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

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	MIL-STD-129	Marking for Shipment and Storage
*	MIL-STD-147	Palletized Unit Loads
	MIL-STD-810	Environmental Test Methods and Engineering Guidelines

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

* 2.1.2 Other Government documents, drawings and publications. The following Government documents form a part of this specification to the extent specified herein.

Title 49	Code of Federal Regulations

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20401)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

*	ASTM D 257	D-C Resistance or Conductance of Insulating Materials
*	ASTM D 732	Shear Strength of Plastics by Punch Tool
*	ASTM D 790	Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
*	ASTM D 1621	Compressive Properties of Rigid Cellular Plastics
*	ASTM D 1623	Tensile and Tensile Adhesion Properties of Rigid Cellular Plastics
	ASTM D 1674	Polymerizable Embedding Compounds Used for Electrical Insulation
*	ASTM D 3574	Flexible Cellular Materials - Slab, Bonded, and Molded Urethane Foams Testing
*	ASTM D 3951	Standard Practices for Commercial Packaging

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(Application for copies should be addressed to the American Society for Testing and Materials, 1961 Race Street, Philadelphia, Pennsylvania 19103).

* (Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First article. When specified, a sample shall be subjected to a first article inspection (see 4.4 and 6.2).

3.2 Materials. The polyurethane foam compounds covered by this specification shall consist of a polyester-type polyurethane base resin and a separate isocyanate-type curing agent, so formulated as to produce, upon mixing, polyurethane foam of a specified density and compressive strength (see 1.2).

3.2.1 Shelf life. The manufacturer of the foam compound shall certify that the foam compound will meet all of the requirements of this specification after 6 months' storage in the original, unopened containers at a temperature not exceeding 26.7° centigrade (C) (80° Fahrenheit (F)).

3.3 Physical properties of cured foam. The foam compound, when mixed and cured in accordance with the manufacturer's recommendations, shall meet all the physica property requirements of Table I. For the purpose of this specification, the foam compound shall be free foamed; i.e., its volume shall not be restricted during foaming.

3.4 Workmanship. The foam compound ingredients shall be homogeneous and free from foreign materials. Containers shall be clean, uniformly filled, well sealed, and legibly marked.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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TABLE I. Physical properties of cured foam

Property to be tested	Conditioning procedure (See 4.6.2 and Table II)	Property Values for Type I	Type II	Type III	Type IV	Type V	Type IV	Test Method Reference
Density, minimum/ maximum, pounds per cubic foot	A	1.8-2.2	4.5-5.5	7.2-8.8	11.0-13.0	16.5-20	1.9-2.2	ASTM D 3574
Compressive strength at 50 percent deformation, minimum, pounds per square inch, psi	A	8	20	100	275	450	40 1/	ASTM D 1621
	E-1/165	1	2	10	20	40	5 1/	
	E-1/-65	8	20	100	275	450	40 1/	
Tensile strength, minimum, psi	A	10	50	100	150	275	35	ASTM D 1623
	E-1/165	7	15	20	40	70	14	
	E-1/-65	10	50	100	150	275	35	
Flexural strength, minimum, psi	A	8	70	175	275	550	40	ASTM D 790
Shear strength, minimum, psi	A	6	20	50	80	150	35	ASTM D 732
Volume resistivity, minimumm ohm-cm	A	6×10^{14}	6×10^{14}	6×10^{14}	6×10^{14}	6×10^{14}	6×10^{14}	ASTM D 257
	E-1/165	6×10^{14}	6×10^{14}	6×10^{14}	6×10^{14}	6×10^{14}	6×10^{14}	
	E-1/-65	6×10^{14}	6×10^{14}	6×10^{14}	6×10^{14}	6×10^{14}	6×10^{14}	
Fungus resistance	A	No evidence of fungus growth						MIL-STD-810 Method 508

Note 1/ - Compressive strength is measured at 10 percent deformation.

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4.2 Definitions. The definitions of inspection terms shall be in accordance with MIL-STD-109.

4.3 Classification of inspection. The classification of inspection shall be as follows:

* a. First article inspection (see 4.4)

b. Acceptance inspection (see 4.5)

* 4.4 First article inspection.

* 4.4.1 First article sample. A first article sample to be inspected under this document shall be representative of the subsequent production and shall meet all the requirements as specified herein. Unless otherwise specified by the procuring activity, the first article sample shall consist of a resin catalyst two-component system.

* 4.4.2 Sampling. Sampling for first article inspection shall be as specified by the procuring activity.

* 4.4.3 Examinations. The first article examinations shall consist of all the acceptance inspections specified herein.

* 4.4.4 Testing. First article testing to determine compliance with all the requirements of Table I shall be conducted in accordance with the corresponding test method reference specified in Table I.

4.5 Acceptance inspection.

4.5.1 Lot size. Lot size shall not exceed that quantity of foam compound produced in a single blending operation by the same manufacturer and manufactured in accordance with the same specification revision. Each lot shall be composed of raw materials from homogeneous lots.

4.5.2 Sampling.

* 4.5.2.1 Sampling for inspection of containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105 at inspection level I and acceptable quality level of 2.5 percent defective to verify compliance with all requirements regarding fill, closure, marking, and other requirements not requiring tests.

* 4.5.2.2 Sampling for testing of prepared specimens. Unless otherwise specified (see 6.2), testing shall be performed on specimens prepared from the base resin and the curing agent selected at random from the lot in sufficient quantities for casting or molding all the specimens, blocks, or slabs needed for testing. The casting, molding, and preparation of specimens shall be in accordance with ASTM D 1674. The number of specimens required for each test shall be in accordance with the number called out in the corresponding test method reference specified in Table I.

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4.5.3 Examinations. Examinations shall be conducted for the purpose of determining compliance with the requirements for shelf life, workmanship, preservation, packaging, packing, and marking.

4.5.3.1 Inspection of containers. Each sample-filled container shall be examined for defects in construction of the container and the closure, for evidence of leakage, and for unsatisfactory marking. Each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects or containing less than the required fill shall be rejected. If the number of defective containers in any sample exceeds the acceptance number for the specified sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.5.4 Testing. Testing to determine compliance with the requirements of Table I shall be conducted in accordance with the corresponding test method reference specified in Table I.

* 4.5.5 Waiver of acceptance inspection testing. Acceptance inspection testing may be waived at the option of the contracting officer, thereby reducing cost and expediting delivery, if a complete inspection of a first article sample has been made and accepted, and each subsequent lot is accompanied by a certificate stating that the lot is of the same type, made of the same composition of materials and formulated by the same process as was the first article sample. The certificate shall be signed by a responsible agent of the contractor and show evidence of this agent's authority to bind the principal.

4.6 Test methods and procedures.

4.6.1 Test specimens. Test specimens shall be prepared as specified in 4.5.2.2. Testing shall be started after conditioning is completed.

4.6.2 Conditioning. Specimens shall be conditioned as specified in Tables I and II prior to testing.

4.6.2.1 Designation. The type of conditioning required shall be designated as follows:

Condition A: As received; no special conditioning.

Condition E: Temperature conditioning.

4.6.2.2 Procedures. The conditioning procedure required, with the exception of condition A, shall be indicated by the following combination of symbols:

a. A capital letter indicating the type of conditioning.

b. A number indicating the duration of the conditioning in hours.

c. A number indicating the conditioning temperature in degrees Fahrenheit.

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The numbers shall be separated from each other by a slash mark and from the capital letter by a dash. A sequence of conditions shall be denoted by the use of a plus (+) sign between successive conditions.

4.6.2.3 Conditioning tolerances. Conditioning tolerances and additional testing conditions shall be as specified in Table II.

* 5. PACKAGING

* 5.1 Preservation. Preservation shall be level A, B, or Industrial as specified (see 6.2).

* 5.1.1 Level A.

* 5.1.1.1 Unit packs. The polyester-type polyurethane base resin and the separate isocyanate-type curing agent shall be unit packaged in accordance with Method IA-5 of MIL-P-116. The polyester-type polyurethane base resin and the isocyanate-type curing agent, in their respective containers, shall be packaged as one unit into a fiberboard box conforming to PPP-B-636, class weather-resistant.

* 5.1.1.2 Intermediate packs. Items, unit packed as specified in 5.1.1.1, shall be placed in intermediate containers conforming to PPP-B-636, class weather-resistant. Intermediate containers shall be uniform in size, shape and quantities, shall be of minimum tare and cube and shall contain multiples of five unit packs, not to exceed 100 unit packs. No intermediate packs are required when the total quantity shipped to a single destination is less than 100 unit packs.

* 5.1.2 Level B. The level B preservation shall conform to the MIL-P-14232 requirements for this level.

* 5.1.3 Industrial. The industrial preservation shall be in accordance with the requirements of ASTM D 3951.

* 5.2 Packing. Packing shall be level A, B, or Industrial, as specified (see 6.2).

* 5.2.1 Level A. The packaged items shall be packed in containers conforming to PPP-B-601, overseas type; PPP-B-621, Class 2, style 4 or PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with applicable container specifications except the strapping shall conform to QQ-S-781, type I or IV, finish A. When the gross weight exceeds 200 pounds or the container length and width is 48" x 24" or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification.

* 5.2.2 Level B. The packaged items shall be packed in fiberboard containers conforming to PPP-B-636, class weather-resistant. Closures shall be in accordance with the appendix thereto.

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TABLE II. Conditioning tolerances and testing conditions

Conditioning Procedure	Conditioning tolerances		Testing conditions	
	Time	Temperature	Temperature and Tolerance	Remarks
E (low temperature)	-0, +6 minutes	± 5 degrees F	-65 ± 5 degrees F	Start test immediately after the cycling has been completed.
E (high temperature)	-0, + 6 minutes	± 5 degrees F	165 ± 5 degrees F	Start test immediately after the cycling has been completed.

* 5.2.3 Industrial. The packaged items shall be packed in accordance with requirements of ASTM D 3951.

* 5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quanities to the greatest extent practicable.

* 5.2.4.1 Level A. Items, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

* 5.2.4.2 Level B. Items, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic.

* 5.3 Marking.

* 5.3.1 Levels A and B. In addition to any special or other identification marking required by the contract (see 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

* 5.3.2 Industrial. Industrial marking shall be in accordance with the requirements of ASTM D 3951.

* 5.4 General.

* 5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

* 5.4.2 Hazardous materials. Packaging, marking, and labeling of hazardous materials shall conform to the requirements of the Code of Federal Regulations, Title 49.

* 6. NOTES

6.1 Intended use. The foam compound covered by this specification is primarily intended for use in potting and encapsulating electronic components to protect them from thermal and mechanical shock (see 6.3).

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

* b. Military part number denoting type and quantity required (see 1.2 and 1.3).

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* c. Whether or not a first article sample is required (see 3.1).

* d. Sampling requirements for inspection of first article sample, as required (see 4.4.2).

e. Selection of material for testing, if other than specified in 4.5.2.2.

* f. Selection of applicable levels of preservation, packaging, packing and marking (see 5.1, 5.2, and 5.3).

6.3 Engineering information.

* 6.3.1 Service temperature range. The foam compounds covered by this specification are suitable for use throughout the temperature range of -54° to +74°C (-65° to +165°F) within the physical limitations of the foam compounds as indicated in Table I.

* 6.4 Asterisks. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-46847B	Plastic Material, Foamed Polyurethane for Encapsulating Electronic Components

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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